



CTU

CZECH TECHNICAL
UNIVERSITY
IN PRAGUE

ENHANCING FAKE NEWS CLASSIFICATION WITH ADVANCED NLP MODELS

Nikita Bardatskii



Goals

- Understand the previous work of Jan Flajžík about fake news detection
- Research new NLP methods to enhance classification on the provided datasets
- Make experiments using advanced NLP models (BERT, GPT-2, etc.)
- Compare results with the previous work

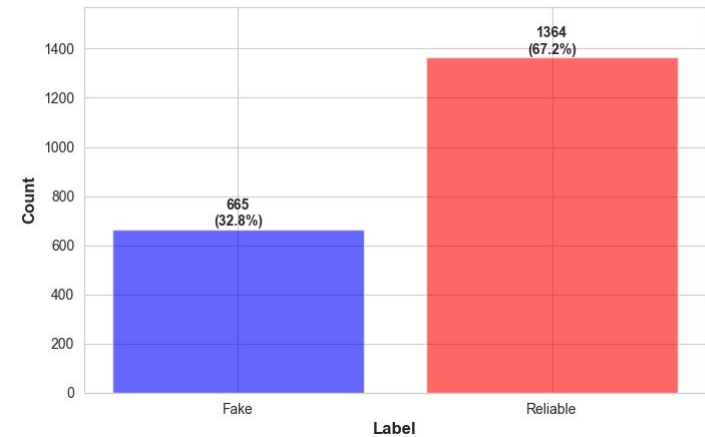
Methodology

- Research latest papers about fake news detection
- Choose more advanced methods
- Conduct experiments
- Compare results with the previous work
- Discuss overall own contribution compared to the previous work and what can be improved

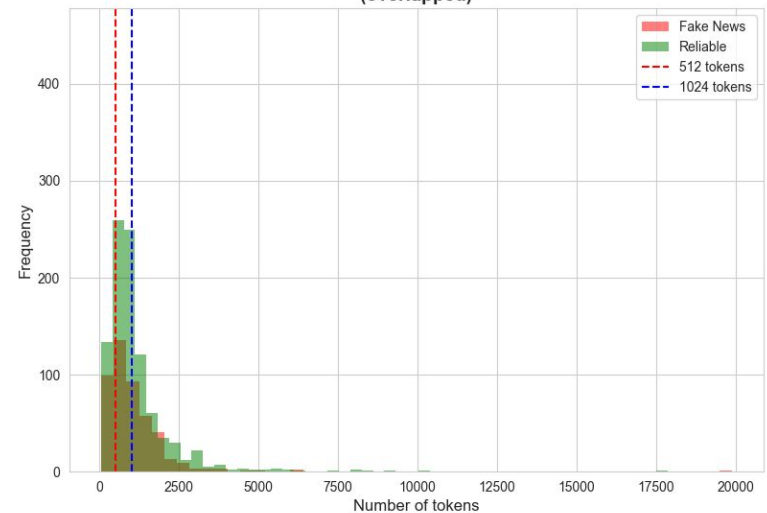
Datasets Analysis

- 3 english, 1 czech datasets
- Long articles, only of the datasets has short summaries
- Advanced visualization of the data
- Tokenization analysis

Label Balance in ReCOVeRY Dataset



Distribution by Label (overlapped)

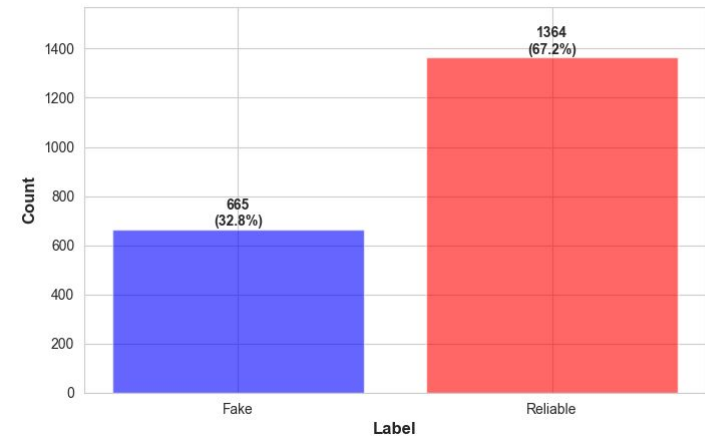


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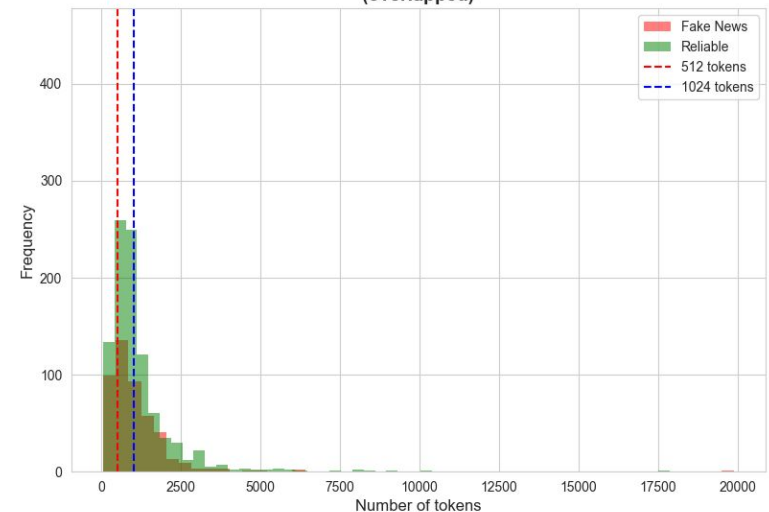
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* Datasets are taken from original thesis

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Distribution by Label (overlapped)



Used Models

- BERT
 - Encoder-only transformer architecture
 - Use bi-directional context (left and right)
 - Baseline model for the classification task

Použité modely

- BERT
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- DistilBERT
 - Simplified version of the BERT created by knowledge distillation
 - Approximately 40 % smaller and 60 % faster
 - Preserve the majority of the BERT performance

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 - Robust, optimized version of the BERT
 - Trained on more data
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- GPT-2
 - Decoder-only transformer architecture
 - Predicts next token based on the previous tokens
 - Primary usage is text generation



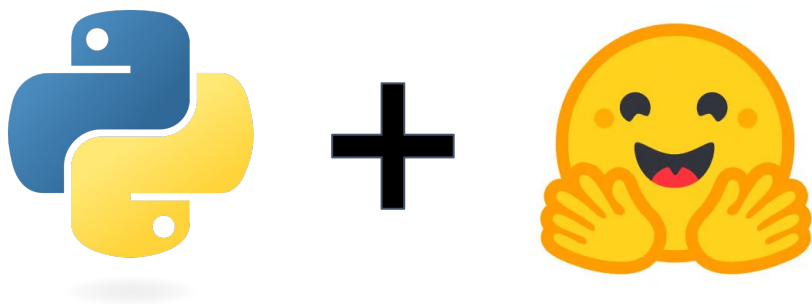
Implementation



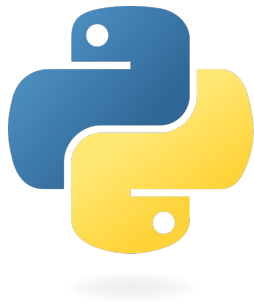
Implementation



Implementation



Implementation



Standard
ML a DS
tools
(scikit, pytorch, scipy,
nltk)

Implementation



+



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Saving and logging

Implementation



+

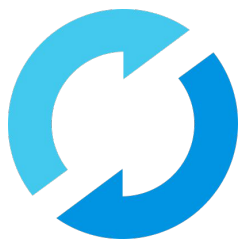


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Standard
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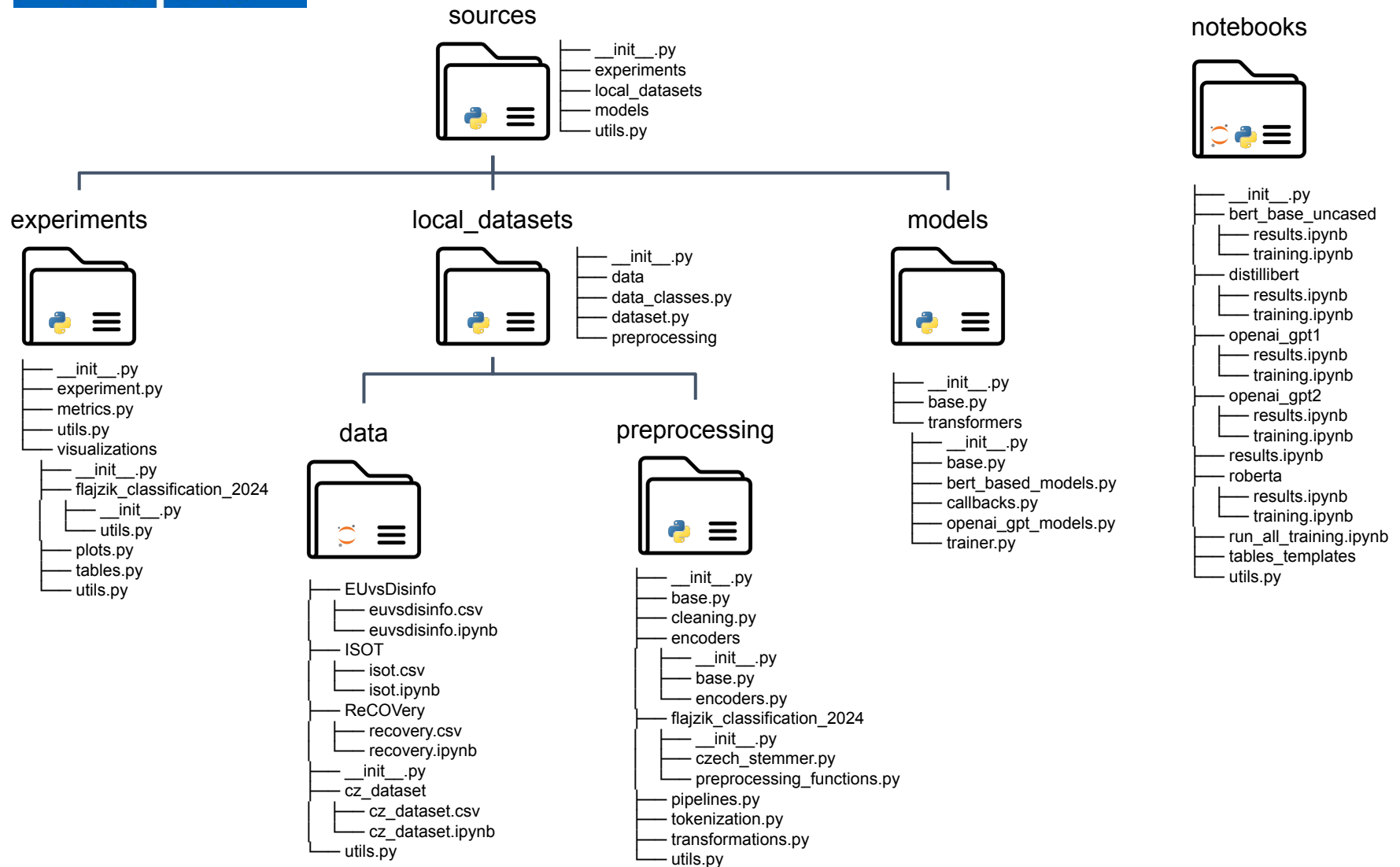
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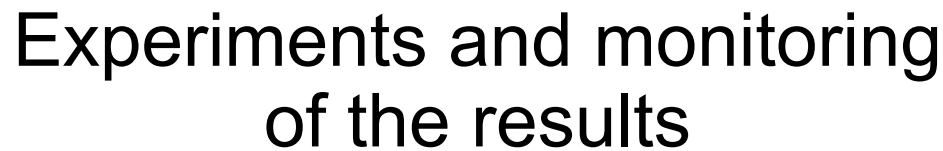


MLflow

Implementation



* assets, images, templates and helper files are excluded



34 matching runs

Presentation of the results

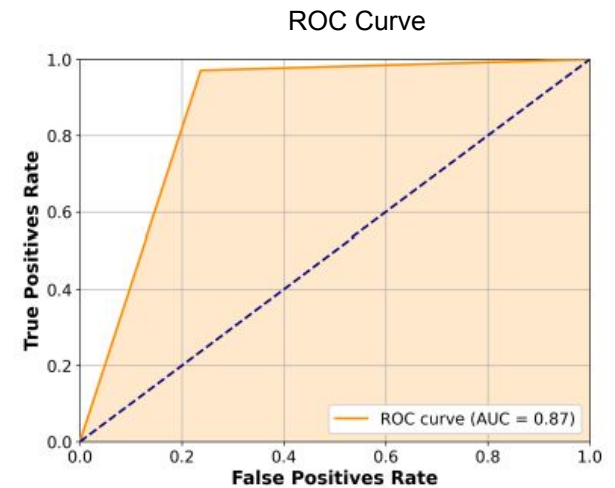
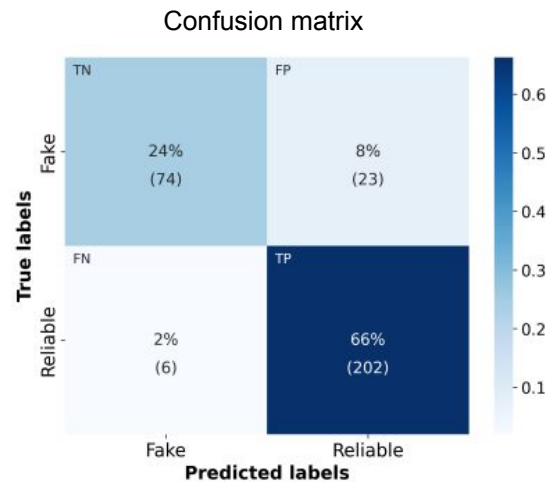
Results for the ReCOVery Dataset:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	FNR
BERT	0.995	0.995	0.995	0.995	1.000	0.006	0.005
DistillIBERT	0.989	0.987	0.991	0.989	0.997	0.014	0.009
RoBERTa	0.989	0.986	0.994	0.990	0.999	0.015	0.006
GPT1	0.987	0.990	0.985	0.988	0.999	0.012	0.015
GPT2	0.990	0.989	0.991	0.990	0.999	0.011	0.009

Results for the best model BERT:

Hyperparameter	Value
epochs	10
batch_size	16
eval_batch_size	16
learning_rate	0.000
lr_scheduler	linear
weight_decay	0.000
optimizer	adamw_torch
early_stopping_patience	3
early_stopping_threshold	0.010

Parameter	Value
pipeline_name	minimal
step_0	bert_base_uncased_encoder



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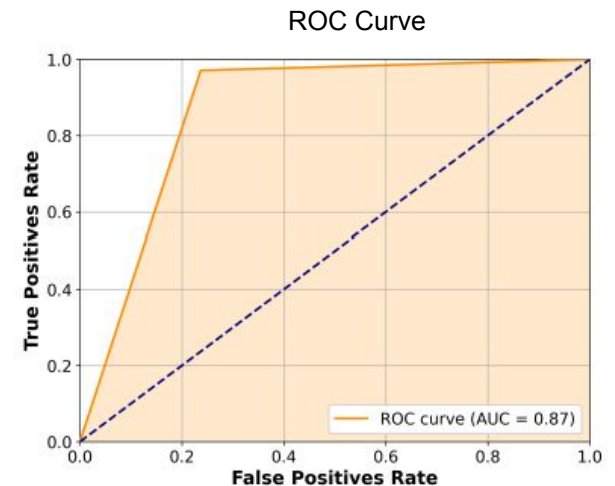
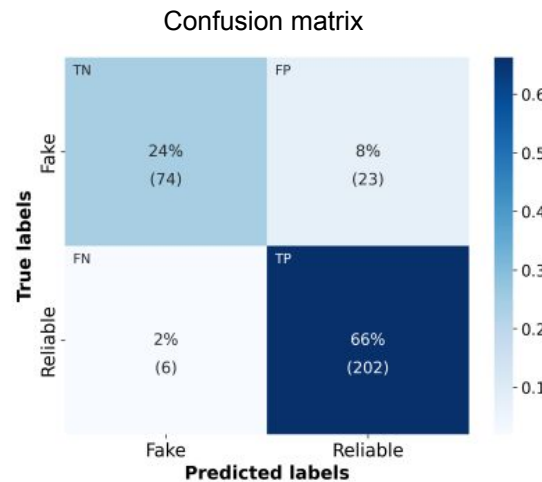
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* hyperparameters tables, ROC Curves and Confusion matrices can be found only in the source code repository

Comparing results with the previous work

ISOT

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.954	0.944	0.966	0.955
Random Forest	0.929	0.991	0.891	0.938
CNN	0.974	0.987	0.965	0.976
LSTM	0.656	0.674	0.686	0.680

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	FNR
BERT	0.995	0.995	0.995	0.995	1.000	0.006	0.005
DistillliBERT	0.989	0.987	0.991	0.989	0.997	0.014	0.009
RoBERTa	0.989	0.986	0.994	0.990	0.999	0.015	0.006
GPT1	0.987	0.990	0.985	0.988	0.999	0.012	0.015
GPT2	0.990	0.989	0.991	0.990	0.999	0.011	0.009

EUvsDisinfo

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.909	0.947	0.875	0.910
Random Forest	0.941	0.936	0.939	0.937
CNN	0.945	0.961	0.935	0.947
LSTM	0.951	0.933	0.966	0.949

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	FNR
BERT	1.000	1.000	1.000	1.000	1.000	0.000	0.000
DistillliBERT	0.978	0.956	1.000	0.977	1.000	0.041	0.000
RoBERTa	0.991	1.000	0.983	0.991	1.000	0.000	0.017
GPT1	0.984	1.000	0.973	0.986	0.999	0.000	0.027
GPT2	0.981	0.982	0.982	0.982	0.998	0.019	0.018

ReCOVary

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.835	0.941	0.834	0.884
Random Forest	0.794	0.956	0.781	0.860
CNN	0.781	0.932	0.779	0.849
LSTM	0.695	0.927	0.709	0.803

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	FNR
BERT	0.905	0.898	0.971	0.933	0.940	0.237	0.029
DistillliBERT	0.866	0.908	0.895	0.901	0.919	0.198	0.105
RoBERTa	0.807	0.818	0.917	0.865	0.850	0.424	0.083
GPT1	0.833	0.880	0.876	0.878	0.889	0.263	0.124
GPT2	0.843	0.886	0.877	0.882	0.912	0.228	0.123

Czech Dataset

Model	Accuracy	Precision	Recall	F1-Score
Naive Bayes	0.899	0.907	0.922	0.915
Random Forest	0.831	1.000	0.778	0.875
CNN	0.801	0.812	0.831	0.821
LSTM	0.749	0.672	0.903	0.771

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	FPR	FNR
BERT	0.847	0.811	0.966	0.882	0.918	0.328	0.034
DistillliBERT	0.920	0.901	0.965	0.932	0.978	0.138	0.035
RoBERTa	0.613	0.613	1.000	0.760	0.775	1.000	0.000
GPT1	0.780	0.787	0.895	0.837	0.826	0.418	0.105
GPT2	0.793	0.787	0.871	0.827	0.865	0.308	0.129

Comparing results with the previous work

- **ISOT:**
 - Best model: **BERT**
 - Accuracy: **90,5% vs 83,5%** with Naive Bayes

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 - Best model: **BERT**
 - Accuracy: **99,5% vs 97%** with CNN

Comparing results with the previous work

- **ISOT:**
 - Best model: **BERT**
 - Accuracy: **90,5% vs 83,5%** with Naive Bayes
- **ReCOVery:**
 - Best model: **BERT**
 - Accuracy: **99,5% vs 97%** with CNN
- **EUvsDisinfo:**
 - Best model: **BERT**
 - Accuracy: **100% vs 95,1%** with LSTM

Comparing results with the previous work

- **ISOT:**
 - Best model: **BERT**
 - Accuracy: **90,5% vs 83,5%** with Naive Bayes
- **ReCOVery:**
 - Best model: **BERT**
 - Accuracy: **99,5% vs 97%** with CNN
- **EUvsDisinfo:**
 - Best model: **BERT**
 - Accuracy: **100% vs 95,1%** with LSTM
- **Czech Dataset:**
 - Best model: **DistilBERT**
 - Accuracy: **92% vs 89,9%** with Naive Bayes

Conclusions

- Experiments with advanced models were conducted and compared to the predecessor's work results
- Some insights on how to improve performance of the models were taken:
 - Solve the problem with limited input
 - Focus on smaller models
 - Try Mixture of Experts
 - Try Claim extraction + Retrieval + Evidence ranking
 - Try other, "harder" datasets
 - Focus on usage in open source a control of alignment.
- Framework for conducting experiments is created, it needs only an extension of models and preprocessing methods.

Thank you for attention